

SpeechTokenizer Codec-only Evaluation Report

SpeechTokenizer Codec-only Evaluation with Latent Perturbation

Generated from runs/codec_only/{valid_20_noise,test_100_noise}

Scope:

- JSCC encoder/decoder used: false
- Wireless channel used: false
- Rayleigh/AWGN/jammer/JSR/effective SINR used: false
- Noise is latent-domain Gaussian perturbation only

Commands:

python eval_codec_only.py --config configs/train_speechtokenizer.yaml --manifest manifests/mini_librispeech/valid.jsonl --split valid --max_items 20 --batch_size 2 --enable_latent_noise_sweep true --latent_noise_snr_db 30 20 10 5 0 --output_dir runs/codec_only/valid_20_noise

python eval_codec_only.py --config configs/train_speechtokenizer.yaml --manifest manifests/mini_librispeech/test.jsonl --split test --max_items 100 --batch_size 4 --enable_latent_noise_sweep true --latent_noise_snr_db 30 20 10 5 0 --output_dir runs/codec_only/test_100_noise

Clean Codec Summary

Run: valid_20_noise

manifest: manifests/mini_librispeech/valid.jsonl

evaluated items: 20

codec type: speechtokenizer

representation shape: [8, 50, 1024]

sample_rate: 16000, waveform_samples: 16000

mean waveform SNR dB: 5.2287

mean SI-SDR dB: 3.8519

mean STFT L1: 0.1078

Run: test_100_noise

manifest: manifests/mini_librispeech/test.jsonl

evaluated items: 100

codec type: speechtokenizer

representation shape: [8, 50, 1024]

sample_rate: 16000, waveform_samples: 16000

mean waveform SNR dB: 4.5160

mean SI-SDR dB: 2.8812

mean STFT L1: 0.0586

Latent Noise Sweep - valid_20_noise

Latent Noise Sweep - valid_20_noise

latent_noise_snr_db is latent-domain perturbation SNR, not wireless SNR.

SNR(dB)	mean SI-SDR	mean waveform SNR	mean STFT L1	dSI-SDR	dWaveSNR	dSTFT
30	3.8511	5.2286	0.1078	-0.0008	-0.0001	0.0000
20	3.8664	5.2389	0.1079	0.0145	0.0102	0.0001
10	3.7875	5.1743	0.1083	-0.0644	-0.0544	0.0005
5	3.6119	5.0279	0.1106	-0.2400	-0.2007	0.0028
0	2.8898	4.5041	0.1151	-0.9621	-0.7246	0.0073

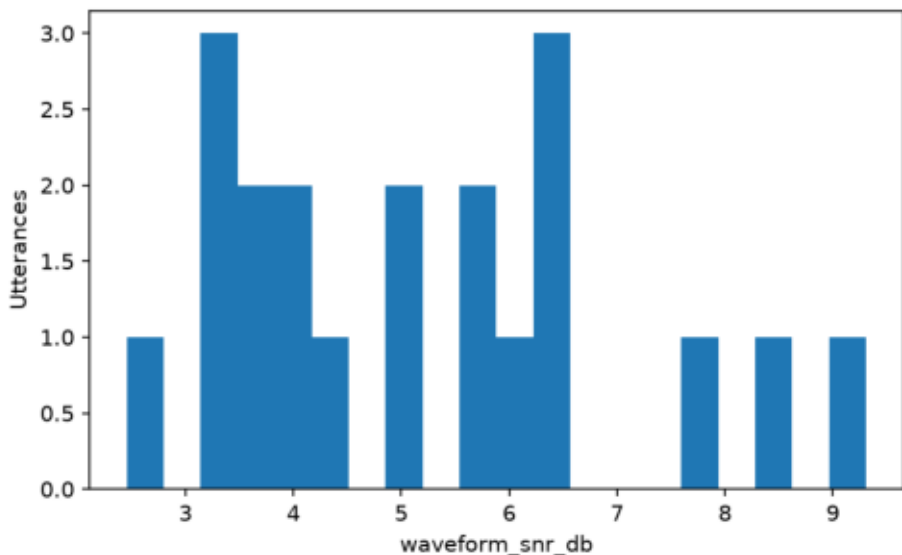
Mean per-layer latent energy:

```
[
  12.471030712127686,
  2.3194423854351043,
  1.4701753735542298,
  1.1253543853759767,
  0.8804731130599975,
  0.7371024012565612,
  0.6388751149177552,
  0.5360215410590172
]
```

Clean Codec Plots - valid_20_noise

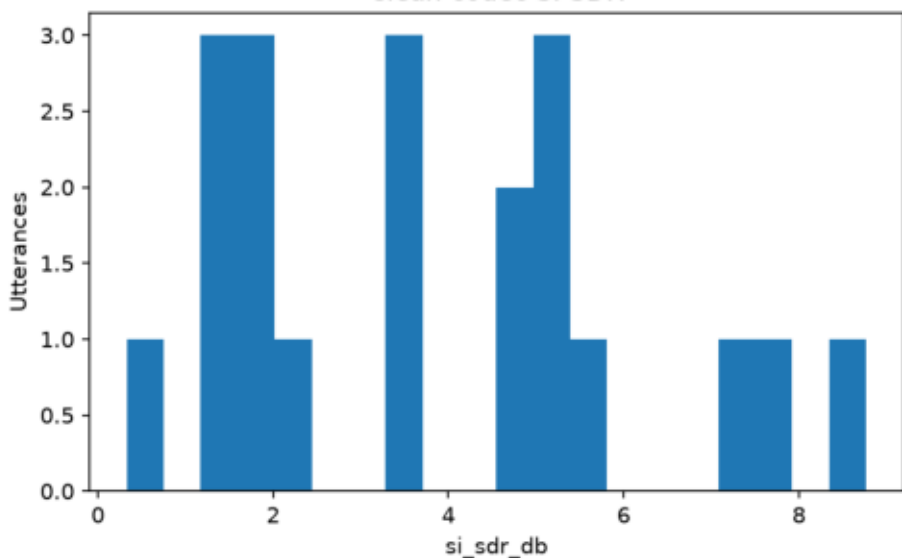
waveform_snr_hist.png

Clean codec waveform SNR



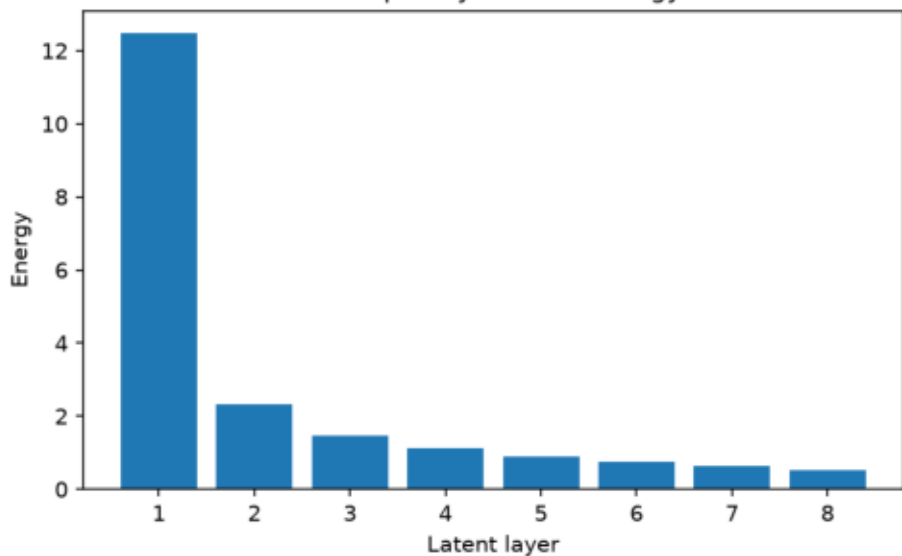
si_sdr_hist.png

Clean codec SI-SDR



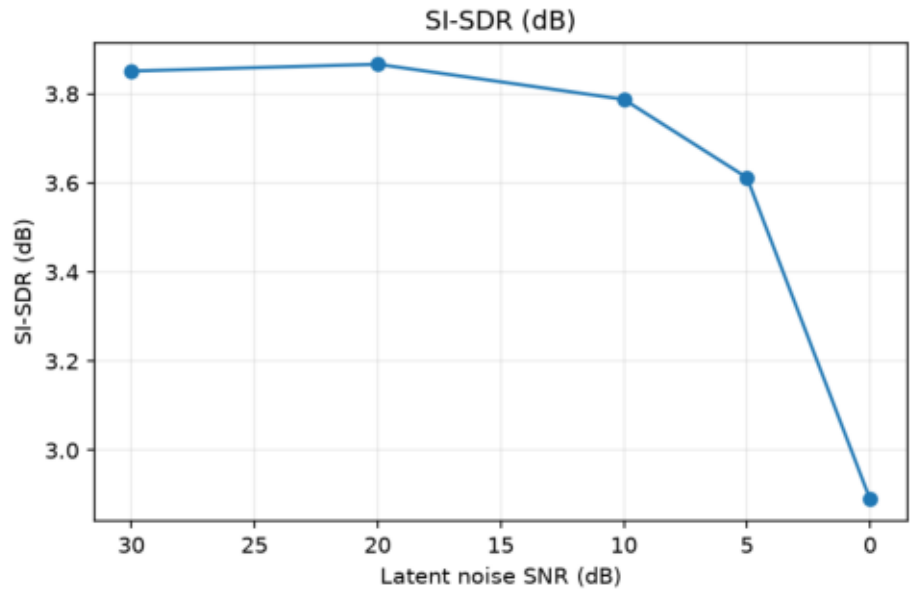
per_layer_energy.png

Mean per-layer latent energy

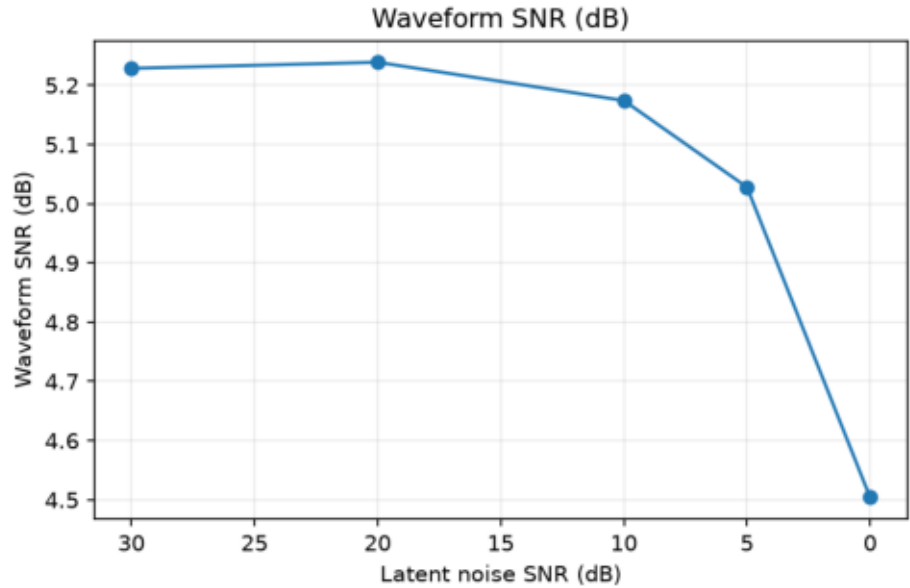


Latent Noise Plots - valid_20_noise

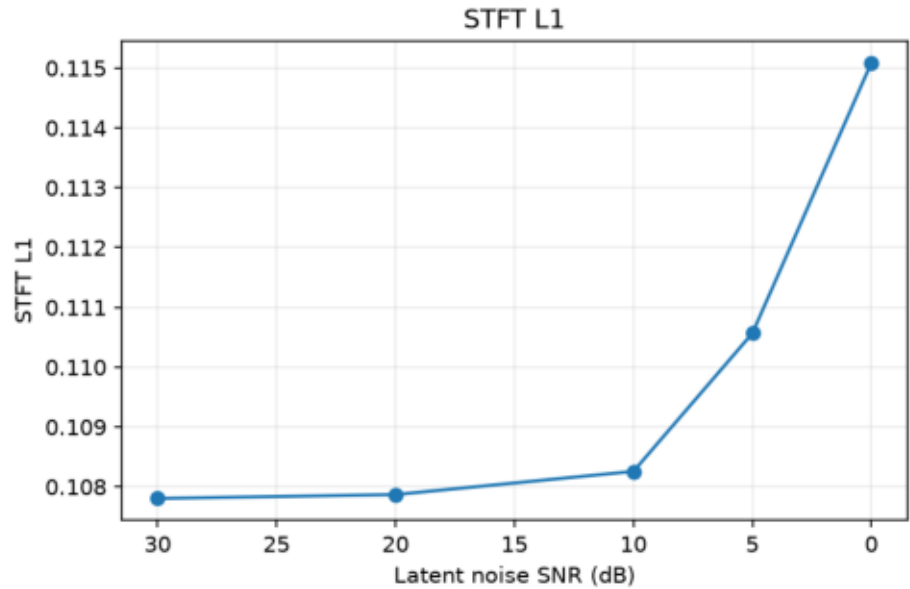
latent_noise_snr_vs_si_sdr.png



latent_noise_snr_vs_waveform_snr.png



latent_noise_snr_vs_stft_l1.png



Latent Noise Sweep - test_100_noise

Latent Noise Sweep - test_100_noise

latent_noise_snr_db is latent-domain perturbation SNR, not wireless SNR.

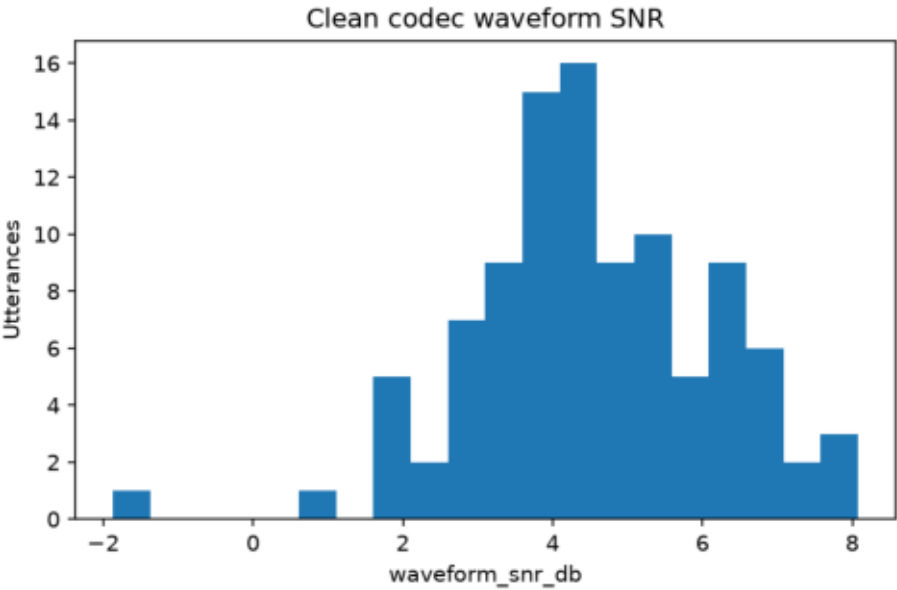
SNR(dB)	mean SI-SDR	mean waveform SNR	mean STFT L1	dSI-SDR	dWaveSNR	dSTFT
30	2.8744	4.5114	0.0586	-0.0068	-0.0045	0.0000
20	2.8660	4.5011	0.0587	-0.0152	-0.0148	0.0001
10	2.7831	4.4420	0.0592	-0.0981	-0.0739	0.0006
5	2.5233	4.2313	0.0606	-0.3579	-0.2847	0.0020
0	1.7840	3.7161	0.0647	-1.0972	-0.7999	0.0061

Mean per-layer latent energy:

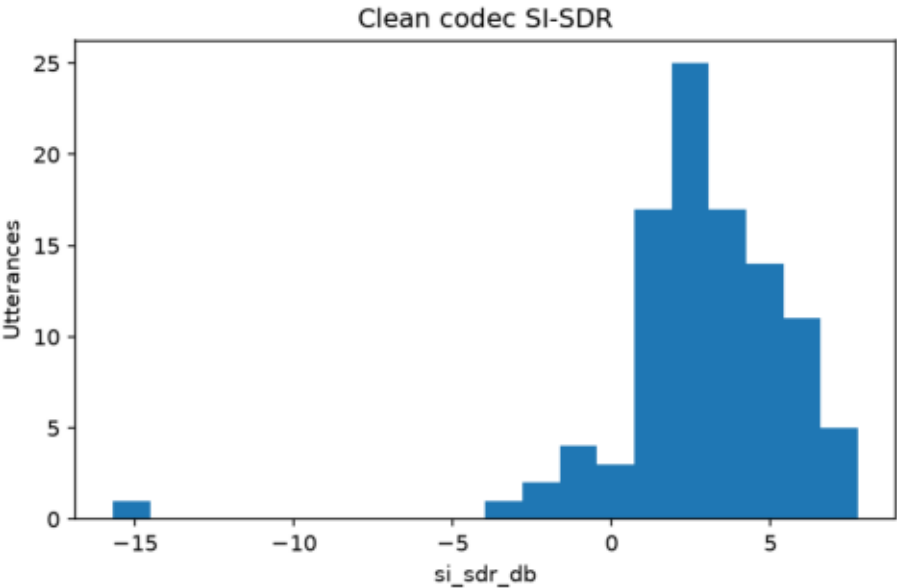
```
[
  12.469777421951294,
  2.2222035908699036,
  1.4747791785001754,
  1.0899338859319687,
  0.8549855303764343,
  0.7173266273736953,
  0.6049509197473526,
  0.526217301785946
]
```

Clean Codec Plots - test_100_noise

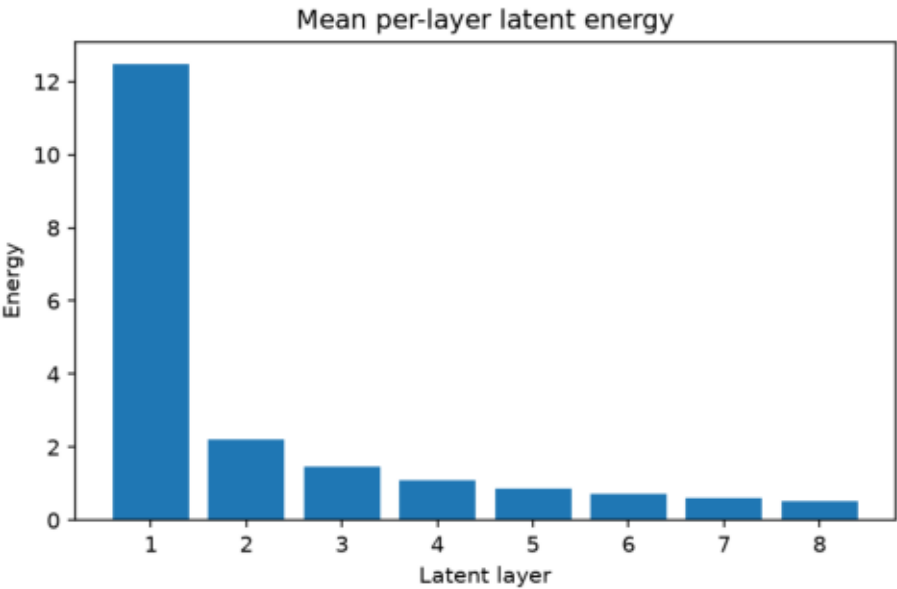
waveform_snr_hist.png



si_sdr_hist.png

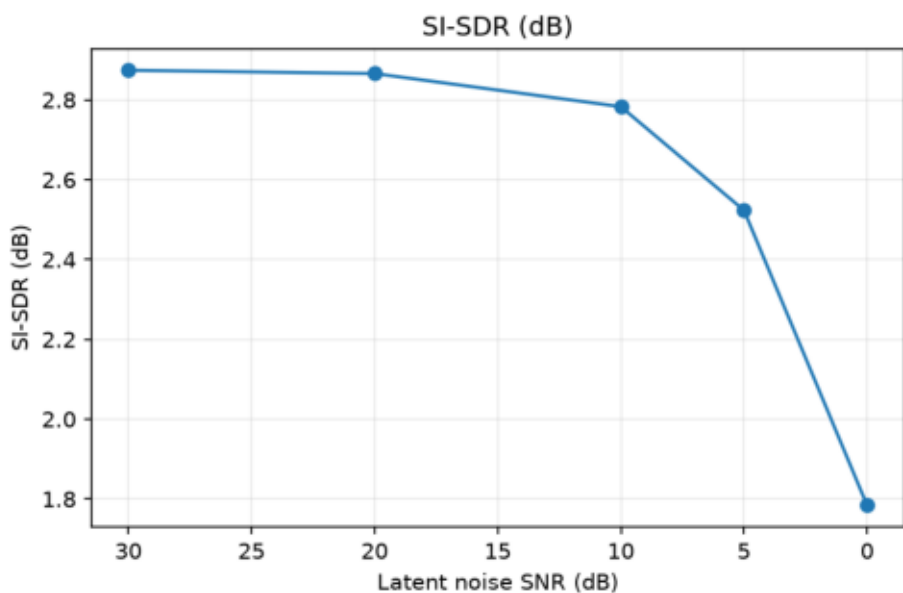


per_layer_energy.png

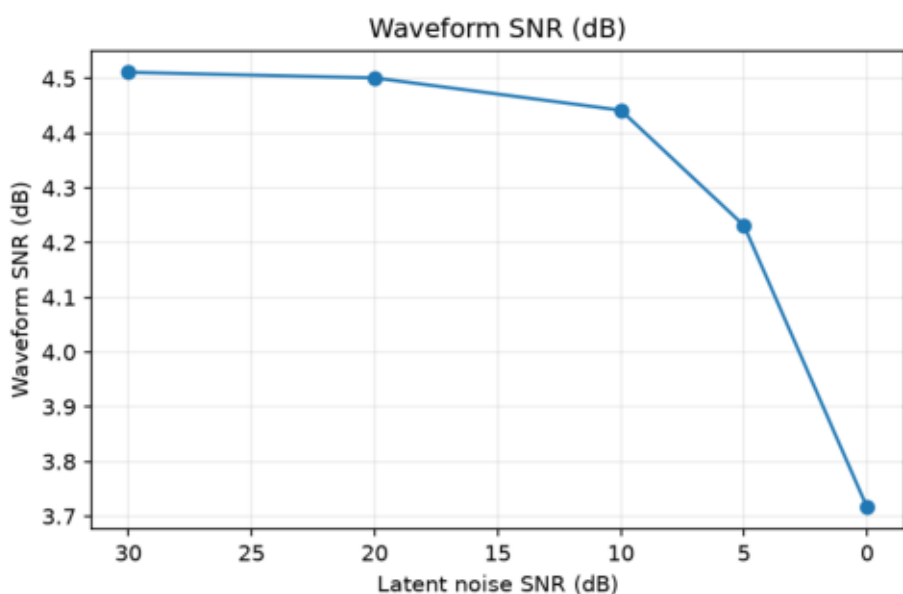


Latent Noise Plots - test_100_noise

latent_noise_snr_vs_si_sdr.png



latent_noise_snr_vs_waveform_snr.png



latent_noise_snr_vs_stft_l1.png

